

Technical Supplement: When Learning Paths Lie – Gradient Retention under Hidden Relations

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Environment Details

510K Card Game

510K is a four-player trick-taking game with a standard 52-card deck. Cards 5, 10, K carry point values (5, 10, 10). The lead player sets the pattern (single, pair, straight, bomb, 510K pattern, etc.); others must follow with a higher-rank play or pass. The trick winner collects all 5/10/K points from the trick.

Action space: 300 discrete actions (0 = pass, 1–299 = pattern index). **Observation space:** 112-dim Box (116-dim for OBVIOUS mode, with 4-bit teammate indicator). **15 pattern types:** SINGLE, PAIR, THREE, THREE_ONE, THREE_PAIR, STRAIGHT (5+), CONSECUTIVE_PAIRS (3+), AIRPLANE_NONE, AIRPLANE_SINGLE, AIRPLANE_PAIR, BOMB, NONSUIT_510K, SUIT_510K, RED_A_SINGLE, RED_A_PAIR.

Four modes share identical card mechanics but differ in team structure:

- **SINGLE:** solo-competitive. Game ends when first player empties hand. Winner +30, losers –10 each (plus own 510K points).
- **STATIC:** fixed teams (players 0&2 vs. 1&3). Both teammates must finish. Winners $\times 2$ multiplier, losers $\times 1$.
- **DYNAMIC:** teams determined by hidden red-Ace distribution (holders of $A\heartsuit/A\diamondsuit$ form one team). Same scoring as STATIC. Red A is the strongest single; red-A pair is the strongest bomb.
- **OBVIOUS:** identical card mechanics to DYNAMIC, but team membership is observable as a 4-bit observation. This isolates *information structure* from card mechanics.

Behavioral features for path integral. Seven mode-independent features (ϕ_1 – ϕ_7), all with VIF < 2 :

- ϕ_1 **MyScore:** cumulative 510K points, normalized by 150.
- ϕ_2 **MyHandSize:** fraction of hand played, $(13 - |h|)/13$.

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- ϕ_3 **MyStrength:** average card rank, mapped to $[0, 1]$.
- ϕ_4 **TrickScore:** points at stake in current trick, normalized by 100.
- ϕ_5 **PassCount:** consecutive passes in current trick, normalized.
- ϕ_6 **ScoreSpread:** std. of scores across players, normalized by 50.
- ϕ_7 **SuppressionGap:** rank gap to leader’s strongest card, normalized by 12.

Feature ablation: deleting any single feature preserves the path-length ordering SINGLE $>$ STATIC $>$ DYNAMIC (7/7 subsets).

OBVIOUS construction. The OBVIOUS observation is 116-dim (112 base + 4 teammate-indicator bits). Bit $i = 1$ if player i is on the agent’s team. This is the only structural difference from DYNAMIC (112-dim).

Toy Matching Environment

A minimal 2-action contextual bandit that isolates IIGC:

- **State:** 1-dim (constant 0) in HIDDEN; 2-dim (one-hot partner type) in REVEALED.
- **Action:** $a \in \{0, 1\}$. Agent chooses which partner type to match.
- **Partner:** hidden binary type $p \in \{0, 1\}$, fixed per episode.
- **Reward:** +1 if $a = p$, –1 if $a \neq p$.
- **Training:** PPO, 10K steps, 10 seeds per condition.

In HIDDEN mode, the agent receives no information about p . Gradients from episodes where $p = 0$ and $p = 1$ cancel completely ($\kappa \approx 0$). In REVEALED mode, the agent observes p directly and learns the optimal type-matching policy.

Overcooked Environment

Grid-world cooperative cooking game with configurable partner roles:

- **State:** 96-dim observation (grid layout, object positions, inventory).
- **Action:** 6 discrete actions (up/down/left/right/interact/stay).

- **Partner roles:** CHEF (cooks soup) or WAITER (delivers soup).
- **Static mode:** partner role visible as 1-hot observation.
- **Dynamic mode:** partner role hidden, switches every 30 steps.
- **Reward:** sparse +20 per soup delivered.
- **Training:** PPO, 1M steps, 8 parallel envs, 8 seeds per mode.

κ Computation

Pseudocode

1. **Input:** trained policy π_θ , two identifiable configurations A and B , rollout count N , and discount γ .
2. Initialize $g_A \leftarrow \mathbf{0}$ and $g_B \leftarrow \mathbf{0}$.
3. For $i = 1, \dots, N$, collect $\tau_A \leftarrow \text{Rollout}(\pi_\theta, A)$ and $\tau_B \leftarrow \text{Rollout}(\pi_\theta, B)$, then add the corresponding return-weighted score-function gradients to g_A/N and g_B/N .
4. Compute $g_{\text{mixed}} \leftarrow (g_A + g_B)/2$ and $\mathcal{E} \leftarrow (\|g_A\|^2 + \|g_B\|^2)/2$.
5. If $\mathcal{E} = 0$, return zero energy and $\kappa = N/A$; otherwise return $\kappa = \|g_{\text{mixed}}\|^2/\mathcal{E}$.

Implementation Notes

- Gradients are computed via REINFORCE $\nabla_\theta \log \pi(a|o) \cdot R(\tau)$, not via backpropagation through a training loss.
- κ requires only **rollout access** to the policy, not training access. It can be computed post-hoc from any saved checkpoint.
- $N \approx 30$ rollouts per configuration suffices for stable κ estimates.
- For value-based methods (DQN), we substitute the TD-loss gradient $\nabla_\theta (Q_\theta(s, a) - y)^2$ in place of the REINFORCE gradient.
- When $\|g_A\|^2 + \|g_B\|^2 = 0$, report zero gradient energy and $\kappa = N/A$. There is no signal to retain, so a numerical ratio is not identifiable.
- When exactly one conditional gradient is zero, $\kappa = 0.5$ by algebra. This is one-sided gradient energy, not evidence that the gradients are orthogonal.

Proofs of Propositions

Proposition 1 (Pythagorean Decomposition)

Let $g_A, g_B \in \mathbb{R}^d$ be the policy gradients under configurations A and B . Define $g_+ = \frac{g_A + g_B}{2}$ (shared component) and $g_- = \frac{g_A - g_B}{2}$ (differential component).

$$\|g_A\|^2 + \|g_B\|^2 = \|g_+ + g_-\|^2 + \|g_+ - g_-\|^2 \quad (1)$$

$$= (\|g_+\|^2 + \|g_-\|^2 + 2g_+ \cdot g_-) \quad (2)$$

$$+ (\|g_+\|^2 + \|g_-\|^2 - 2g_+ \cdot g_-) \quad (3)$$

$$= 2(\|g_+\|^2 + \|g_-\|^2) \quad (4)$$

Therefore $\frac{\|g_A\|^2 + \|g_B\|^2}{2} = \|g_+\|^2 + \|g_-\|^2$. Since $\kappa = \|g_+\|^2 / (\|g_+\|^2 + \|g_-\|^2)$, it follows that κ is the fraction of total gradient energy retained after aggregation.

Proposition 2 (Effective Learning Signal)

Under balanced configuration mixing ($p_A = p_B = 1/2$), the expected per-step gradient during training is:

$$\mathbb{E}[g_{\text{train}}] = \frac{1}{2}(g_A + g_B) = g_+ \quad (5)$$

The effective gradient magnitude satisfies:

$$\|g_+\|^2 = \kappa \cdot \frac{\|g_A\|^2 + \|g_B\|^2}{2} \quad (6)$$

When $\kappa \rightarrow 0$, $\|g_+\|^2 \rightarrow 0$ and the policy receives no effective learning signal regardless of the per-configuration gradient magnitudes.

Proposition 3 (Partition Refinement)

Let a fine relation partition $\mathcal{R}$ refine a coarse partition $\mathcal{C}$, with $g_c = \mathbb{E}[g_r \mid c]$. Both partitions have the same mixed gradient. For every coarse cell c , Jensen's inequality gives

$$\sum_{r \in c} p(r \mid c) \|g_r\|^2 \geq \left\| \sum_{r \in c} p(r \mid c) g_r \right\|^2 = \|g_c\|^2. \quad (7)$$

Multiplying by p_c and summing over c shows that fine-partition energy is at least coarse-partition energy. Since the retained numerator is unchanged, $\kappa_{\mathcal{R}} \leq \kappa_{\mathcal{C}}$ whenever both ratios are defined.

Proposition 4 (Gradient-Field Specificity)

For policy-gradient methods, g_r^{PG} reflects the direction of improvement in action preferences for configuration r . A TD-loss gradient instead reflects prediction residuals under its data distribution and target.

Consequently there is no estimator-independent ordering between $\kappa(g_A^{\text{PG}}, g_B^{\text{PG}})$ and $\kappa(g_A^{\text{TD}}, g_B^{\text{TD}})$. Changing the update field can change both the angle and the norm ratio of the conditional vectors even for the same environment and policy. Cross-algorithm values must therefore name the estimator; an empirical reversal is not a proof that either family is immune to hidden information.

Full Path Integral Data

Policy (seed)	Path Length	Endpoint Dist.	Curvature
SINGLE (41)	0.340	0.046	$7.3\times$
SINGLE (51)	0.482	0.068	$7.1\times$
SINGLE (52)	0.561	0.034	$16.6\times$
SINGLE (53)	0.454	0.108	$4.2\times$
SINGLE (54)	0.443	0.051	$8.8\times$
STATIC (41)	0.253	0.071	$3.6\times$
STATIC (61)	0.342	0.086	$4.0\times$
STATIC (62)	0.490	0.073	$6.7\times$
STATIC (63)	0.281	0.048	$5.9\times$
STATIC (64)	0.278	0.021	$13.5\times$
DYNAMIC (41)	0.299	0.046	$6.5\times$
DYNAMIC (42)	0.231	0.112	$2.1\times$
DYNAMIC (43)	0.367	0.026	$14.3\times$
DYNAMIC (44)	0.274	0.031	$8.9\times$
OBVIOUS (71)	0.203	0.100	$2.0\times$
OBVIOUS (72)	0.355	0.122	$2.9\times$
OBVIOUS (73)	0.366	0.051	$7.2\times$
OBVIOUS (74)	0.302	0.022	$13.6\times$
OBVIOUS (75)	0.318	0.036	$8.9\times$
OBVIOUS (76)	0.391	0.085	$4.6\times$
OBVIOUS (77)	0.283	0.043	$6.6\times$
OBVIOUS (78)	0.404	0.035	$11.5\times$

Table 1: Full path integral data for all 22 seeds across 4 modes.

Extended κ Data

Env	Alg.	Mode	Retention (mean $\pm$ std)	n
Toy	PPO	HIDDEN	0.039 ± 0.073	10
Toy	PPO	REVEALED	0.726 ± 0.100	10
Toy	A2C	HIDDEN	0.195 ± 0.352	10
Toy	A2C	REVEALED	0.842 ± 0.025	10
510K	A2C	SINGLE	0.644 ± 0.201	8
510K	A2C	STATIC	0.500 ± 0.000	8
510K	A2C	DYNAMIC	0.519 ± 0.060	8
510K	DQN	SINGLE	0.797 ± 0.123	8
510K	DQN	DYNAMIC	0.917 ± 0.064	8
510K	SAC	SINGLE	0.504 ± 0.038	2
510K	SAC	DYNAMIC	0.540 ± 0.069	2
510K	PPO	SINGLE	0.569	1
510K	PPO	DYNAMIC	0.444 ± 0.069	9
510K	REINFORCE	SINGLE	0.487 ± 0.312	8
510K	REINFORCE	DYNAMIC	0.605 ± 0.238	8
Overcooked	PPO	STATIC	0.500 ± 0.006	8
Overcooked	PPO	DYNAMIC	N/A (zero energy)	8
Overcooked	A2C	STATIC	0.500 ± 0.000	8
Overcooked	A2C	DYNAMIC	0.125 ± 0.217	8
Overcooked	DQN	STATIC	0.473 ± 0.093	8
Overcooked	DQN	DYNAMIC	0.645 ± 0.209	8

Table 2: Complete stored dataset across environments, algorithms, and modes. The PPO Overcooked DYNAMIC cell has zero conditional-gradient energy, so its retention ratio is undefined rather than evidence of cancellation.

Split-Batch Reveal Audit

Reveal Fraction	Split-batch ratio (mean $\pm$ std)	n
0%	0.534 ± 0.038	2
25%	0.446 ± 0.085	2
50%	0.514 ± 0.024	2
75%	0.324 ± 0.079	2
100%	0.616 ± 0.039	2

Table 3: Two independent rollout batches under each reveal condition (510K, PPO). These ratios audit sampling consistency; because the batches do not index distinct latent relations, they are not relation-conditioned retention estimates.

Training Hyperparameters

Parameter	Value
Learning rate	3×10^{-4}
PPO n_steps	2048 (self-play) / 256 (parallel)
Batch size	64 / 256
Epochs per update	10
γ (discount)	0.99
GAE λ	0.95
Clip range	0.2
Entropy coeff.	0.01
Network	MLP [256, 256], tanh activation
A2C n_steps	256
DQN buffer size	50,000
DQN learning starts	5,000
DQN τ (target update)	0.005
DQN exploration	ϵ -greedy: $1.0 \rightarrow 0.02$ over 30% of training
SAC α	Auto-tuned (initial 0.2)
REINFORCE lr	1×10^{-3}

Table 4: Training hyperparameters for all algorithms.

Computing Infrastructure

Hardware: NVIDIA GeForce RTX 5070 Ti Laptop GPU (12 GB VRAM), Intel(R) UHD Graphics (auxiliary), and 32 GB RAM. **Software:** Windows 11; PyTorch 2.x; stable-baselines3 and sb3-contrib 2.x; Gymnasium 0.29+; NumPy 1.24+.

Code Availability

All source code is publicly available under the MIT License:

- **Main repository:** <https://github.com/flavieninpekin/Information-Induced-Gradient-Contraction-in-Partially-Observable-MARL>
- **Reproduction package:** <https://github.com/flavieninpekin/AAAI2027-510k-clear>
- **510K environment:** https://github.com/flavieninpekin/510k_env

The reproduction package contains pre-computed validation results, all experiment configurations, and the entry point experiments/reproduce_all.py.